

Response submitted by Peru

Questionnaire on the Application of International Law in OAS Member States in the Cyber Context

Questions 2 to 10

Question 2. Do existing fields of international law (including the prohibition on the use of force, the right of self-defense, international humanitarian law, and human rights) apply to cyberspace? Are there areas where the novelty of cyberspace excludes the application of a particular set of international legal rights or obligations?

In first place, it should be recalled that the Charter of the United Nations provides that states shall refrain from the use of force *“against the territorial integrity or political independence of any state, or in any other manner inconsistent with the Purposes of the United Nations”* (Article 2.4), inasmuch as, according to Article 1.1 of the Charter, those purposes are to maintain international peace and security, forswear acts of aggression, and settle disputes by peaceful means.

In addition, Article 1.3 of the UN Charter indicates that the purpose of cooperation among countries is to solve international problems, *inter alia*, of a humanitarian character, and to encourage respect for human rights and for fundamental freedoms for all. Therefore, bearing in mind the fundamental role of the Charter in terms of how it relates to other international instruments (it states at Article 103 that in the event of a conflict with other international agreements, countries' obligations under the Charter shall prevail), it would be reasonable to conclude that no area of international relations lies outside the scope of the aforesaid principles. Bearing in mind that cyberspace is becoming an everyday setting for international interaction, the actors in such interactions are required to observe their higher obligations under international law, including the prohibition on the use of force, the right of self-defense, and respect for human rights and international humanitarian law.

In the sphere of human rights, sensitive issues such as the right to privacy, freedom of information, freedom of expression, free and equal access to information, elimination of the digital divide, intellectual property rights, free flow of information, the right to confidentiality of communications, etc., are entirely valid in cyberspace. Furthermore, in resolution A/HRC/20/L.13 of June 19, 2012, the United Nations Human Rights Council noted that *“the exercise of human rights, in particular the right to freedom of expression, on the Internet is an issue of increasing interest”*; therefore, such rights should be protected in that space.

In sum, the aforementioned areas, as a part of current international law, are applicable to cyberspace, which is nothing more than a juridical variable of a technological nature that amounts to a new context that, to be sure, presents challenges in terms of the application and enrichment of international law, but does not constitute, *per se*, an impediment to the validity of its principles and norms.

Question 3. Can a cyber operation by itself constitute a use of force? Can it constitute an armed attack that triggers a right of self-defense under Article 51 of the UN Charter? Can a cyber operation qualify as a use of force or armed attack

without causing the violent effects that have been used to mark such thresholds in past kinetic conflicts?

The definition of use of force is closely connected with that of armed conflict, which may be international or noninternational in nature. One speaks of an international armed conflict when the confrontation is between the armed forces of two or more States, and noninternational when the confrontation is between the armed forces of a State and nonstate insurgent forces.

In addition, to qualify as an armed conflict, the use of force in combat has to meet two conditions, according to the International Criminal Tribunal for the former Yugoslavia: (1) The hostilities must have a level of intensity that exceeds that of isolated acts; and (2) the parties must have a level of organization that allows coordinated actions to be carried out (See Pinto, Mónica (2003), *La noción de conflicto armado en la jurisprudencia del Tribunal Penal Internacional para la ex-Yugoslavia* [The notion of armed conflict in the jurisprudence of the International Criminal Tribunal for the former Yugoslavia]).

United Nations General Assembly resolution 3314 (XXIX) of 14 December 1974, defines *aggression*—namely, illegitimate use of force under international law—as “*the use of armed force by a State against the sovereignty, territorial integrity or political independence of another State, or in any other manner inconsistent with the Charter of the United Nations.*”

Consequently, for the use of lethal force by means of the machinery of warfare to constitute an act of aggression against a State, it must violate its:

- i. Sovereignty,
- ii. territorial integrity, and/or
- iii. political independence.

On this point, it would be appropriate first to define cyberspace, cyber attack, and cyber warfare before analyzing a possible act of aggression through a cyber operation.

Cyberspace: The place that is not physical, but one of virtual reality, in which electronic communication, especially via the Internet, occurs. While an isolated virtual space cannot constitute part of cyberspace, its configuration requires the concurrence of interdependent networks of information technology infrastructures (see Eichensehr, Kristen E. (2015) *The Cyber-Law of Nations*, The Georgetown Law Journal, Vol. 103, pp. 323 and 324).

Cybernetic attack or cyber attack: It is necessary to distinguish a cyber attack from an abrupt disruption of communications in cyberspace (cyber disruption). Cyber disruptions are cyber operations that cause inconvenience, even extreme inconvenience, but no direct injury or death, and no destruction of property; they are confined to a level in which communications are only disrupted or computer data destroyed. A cyber attack, on the other hand, entails using cyberspace as a means to cause damage to a militarily significant target, resulting in its total or partial destruction, capture, or neutralization (see Brown, Gary D. (2017) *International Law applies to Cyber Warfare! Now What?*, pp. 366 to 368).

Cyber warfare: acts carried out in cyberspace with effects comparable to the use of force (*jus ad bellum*), which must be conducted in accordance with the *jus in bello*. It is worth bearing in mind that the International Committee of the Red Cross (ICRC) has

rejected any strict interpretation of "use of force" that would limit the term to the use of kinetic force, allowing non-kinetic operations that generate comparable effects to be classed as a "use of force." (See Schmitt, Michael N. (2014), *The Law of Cyber Warfare: Quo Vadis?*, Stanford Law & Policy Review, Vol. 25, pp. 269 to 279).

Based on the foregoing, one might reasonably say that a cyber operation that entails or causes damage of that order of magnitude to military targets or individuals or property protected by international humanitarian law could in itself constitute a use of force. In that sense, the inherent right of states to legitimate self-defense under Article 51 of the United Nations Charter ought to be considered valid.

With respect to the possibility of a cyber operation that does not cause violent effects being classed as a use of force or an armed attack, it should be noted that if military damage is not caused, the threshold would be met if the act was likely to cause death or injury to persons or property protected from attacks, provided that the military act also satisfies the requirements of direct causation and belligerent nexus. Such direct causation requires the existence of a "close causal relation between the act and the resulting harm," while belligerent nexus implies that "the act must be specifically designed to directly cause the required threshold of harm in support of a party to the conflict and to the detriment of another" (see Melzer, Nils (2010), *Interpretive Guidance on the Notion of Direct Participation in Hostilities under International Humanitarian Law*, International Committee of the Red Cross (ICRC), Geneva, pp. 50, 52, and 59).

Thus, if real-world damage were to occur as a result of a cyber attack that caused a bomb to explode or a missile to be launched, there would be no impediment to classifying such an action as a use of force (see Hongju Koh, Harold (2012), *International Law in Cyberspace*. In: Faculty Scholarship Series, Vol. 54, Yale Law School Faculty Scholarship).

Clearly, then, a cyber operation may constitute a use of force; for that to be the case the cyber operation would not necessarily have to amount to an act of aggression within the definition contained in United Nations General Assembly resolution 3314 (XXIX). As the International Court of Justice has held, it will be necessary to distinguish the most grave forms of the use of force (those constituting an armed attack) from other less grave forms. (International Court of Justice, Case concerning Military and Paramilitary Activities in and against Nicaragua (United States v. Nicaragua), 1986, § 191). Accordingly, a cyber operation could give rise to a less grave use of force and, therefore, breach Article 2.4 of the Charter.

Just as Article 2.4 is applicable to cyberspace, a State that sustains an armed attack can invoke the right of legitimate self-defense under Article 51 of the Charter.

As to whether a cyber operation can qualify as a use of force or an armed attack without causing violent effects, this General Office is of the opinion that a use of force does not necessarily cause violent effects such that it should be regarded as violating Article 2.4 of the Charter. An interesting scenario in that regard is when a state fires a missile through the airspace of a neighboring state without it descending into its territorial space. Such an act does not produce material damage but it does violate the territory of the neighboring state; accordingly, although it has not generated "violent effects," it infringes the principle contained in the aforementioned Article 2.4 nevertheless.

Question 4. Outside of armed conflicts, when would a State be responsible for the cyber operations of a non-State actor? What levels of control or involvement

must a State have with respect to the non-State actor's operations to trigger the international legal responsibility of that State?

It is worth mentioning that, unlike the other spaces where human and interstate relations play out, that is, continental and insular spaces, seas, rivers, lakes, aerospace, and outer space, cyberspace is the only one to have been "created" by humans. This relatively new sphere of social life is characterized by its artificiality and immateriality. Thus, there is great *"uncertainty in attribution, and levels of attribution, of cyber attacks."* In addition, digital connection systems are practically "omnipresent" and the presence of a vast numbers of possible actors in different parts of the globe (and even, potentially, if technology allows it, away from it) makes identification tremendously complicated. (See Wegener, Henning (2011), *A Concept of Cyber Peace*. In: *The Quest for Cyber Peace*, International Telecommunication Union and World Federation of Scientists, p. 93).

In light of the above, the possibility for states to control those who use cyberspace to unleash attacks over the Internet is far more limited than their potential control over nonstate actors that move about in their territories. Be that as it may, United Nations General Assembly resolution AG/56/83 on Responsibility of States for internationally wrongful acts, adopted on 12 December 2001, provides that for the purposes of attributing international responsibility to a State, "[t]he conduct of a person or entity which ... is empowered by the law of that State to exercise elements of the governmental authority shall be considered an act of the State under international law, provided the person or entity is acting in that capacity in the particular instance," or when that *"person or group of persons is in fact acting on the instructions of, or under the direction or control of, that State"* (Article 5 and 8). By that token, the actions a nonstate actor that are completely alien to the State could not engage the international responsibility of the State. However, if there is inertia on the part of a State in controlling a nonstate actor that unleashed a cyber attack against another State, despite having the capacity to control them, then that could give rise to their conduct being attributable to the State, bearing in mind that Article 11 of the aforementioned resolution AG/56/83 provides: *"Conduct which is not attributable to a State ... shall nevertheless be considered an act of that State under international law if and to the extent that the State acknowledges and adopts the conduct in question as its own."*

Question 5. Are the standards of State responsibility the same or different in the context of an armed conflict as that term is defined in Articles 2 and 3 common to the 1949 Geneva Conventions?

Common Article 2 of the Geneva Conventions of 1949, concerning the application of those instruments, provides that they shall apply to all cases of armed conflict which may arise between two or more of the High Contracting Parties; that is, between States. Having said that, it is worth mentioning that the wording of the provision was conceived in territorial terms. It says, for example, that *"[t]he Convention shall also apply to all cases of partial or total occupation of the territory of a High Contracting Party, even if the said occupation meets with no armed resistance"* (Article 2, second paragraph). Likewise, Common Article 3 of the Geneva Conventions of 1949 provides: *"In the case of armed conflict not of an international character occurring in the territory of one of the High Contracting Parties,"* they shall have the obligation to treat those not directly involved in the will hostilities in a humane manner, without any adverse distinction. Thus, it is completely prohibited for such persons to be murdered, tortured, or subjected to cruel treatment; be taken hostage; have outrages committed upon their personal dignity through humiliating and degrading treatment; or have sentences

passed on them without a fair trial. Furthermore, the parties to the conflict are required to collect and care for the wounded and sick.

It should also be clarified that Common Articles 2 and 3 of the Geneva Conventions of 1949 do not enshrine a definition of armed conflict. The former makes reference to “cases of declared war or of any other armed conflict which may arise between two or more of the High Contracting Parties” and to “*cases of partial or total occupation of the territory of a High Contracting Party, even if the said occupation meets with no armed resistance.*” The latter article, for its part, alludes to “*armed conflict not of an international character.*” In that sense, the aforementioned provisions refer only to international armed conflicts (Article 2) and non-international armed conflicts (Article 3), without providing a definition for either.

The question seeks to determine whether the rules on state responsibility are the same, or, given the particular characteristics of an armed conflict, if there could be different rules on state responsibility.

The answer should be based on Article 55 of the Draft Articles on Responsibility of States for Internationally Wrongful Acts, which provides: “These articles do not apply where and to the extent that the conditions for the existence of an internationally wrongful act or the content or implementation of the international responsibility of a State are governed by special rules of international law.”

It is worth pointing out that in the context of the relationship between international human rights law and international humanitarian law the International Court of Justice has put forward an argument that is relevant for the purposes of settling the question posed:

*“More generally, the Court considers that the protection offered by human rights conventions does not cease in case of armed conflict, save through the effect of provisions for derogation of the kind to be found in Article 4 of the International Covenant on Civil and Political Rights. As regards the relationship between international humanitarian law and human rights law, there are thus three possible situations: some rights may be exclusively matters of international humanitarian law; others may be exclusively matters of human rights law; yet others may be matters of both these branches of international law. In order to answer the question put to it, the Court will have to take into consideration both these branches of international law, namely human rights law and, as *lex specialis*, international humanitarian law.”* (International Court of Justice. Legal Consequences of the Construction of a Wall in the Occupied Palestinian Territory, Advisory Opinion, 9 July 2004, par. 106.)

In this passage of its Advisory Opinion, the International Court of Justice indicates that protection and, with it, the application of human rights treaties do not cease in the context of armed conflict unless their provisions state otherwise. Article 55 of the Draft Articles on Responsibility of States for Internationally Wrongful Acts states that those provisions do not apply in the case of *lex specialis*.

An examination of the Geneva Conventions does not disclose any difference with respect to the provisions on international responsibility set down in the Draft Articles on Responsibility of States for Internationally Wrongful Acts; therefore, it cannot be argued that the draft articles have a different scope of application.

Question 6. Under international humanitarian law, can a cyber operation qualify as an “attack” for the rules governing the conduct of hostilities if it does not

cause death, injury or direct physical harm to the targeted computer system or the infrastructure it supports? Could a cyber operation that produces only a loss of functionality, for example, qualify as an attack? If so, in which cases?

Cyber operations can exceed the threshold of mere unfriendly activities on the web and qualify as use of force if people and public or private property are physically harmed as a result of a cyber attack. Context is crucial in making an evaluation in that regard, as is identifying the actors perpetrating the attack, the target, the effects and the intent. Clear examples would be a cyber attack on a nuclear power plant or the control, manipulation, neutralization, and destruction of an enemy arsenal by means of virtual networks (see Hongju Koh, Harold (2012), *International Law in Cyberspace*. In: Faculty Scholarship Series, Vol. 54, Yale Law School Faculty Scholarship).

Question 7. Is a cyber operation that only targets data governed by the international humanitarian law obligation to direct attacks only against military objectives and not against civilian objects?

It is a recognized principle of international humanitarian law that an attack can only target a military objective. Military objectives are defined as those which *“by their nature, location, purpose or use make an effective contribution to military action and whose total or partial destruction, capture or neutralization, in the circumstances ruling at the time, offers a definite military advantage”* (Article 52 of Additional Protocol I to the Geneva Conventions of August 12, 1949, relating to the protection of victims of international armed conflicts, of June 8 1977).

Thus, for international humanitarian law to be considered applicable to the hypothetical case presented, the "data" would have to qualify as a "military objective." On that premise, the data present in cyberspace would have to make an effective contribution to obtaining a military advantage. Possible examples could be software that allows communication between army troops in the field, that synchronizes a country's missile arsenal, or that helps to locate enemy aircraft. In such cases, an attack on the "data" would cause significant military harm to the opposing forces, and therefore they could be considered "military objectives."

It would be a different matter in the case of a data system that enabled the operating room of a field hospital treating war wounded or civilians to function. Rendering it inoperative would not create a legitimate military advantage, and therefore it would be appropriate to regard an attack on it as a violation of international humanitarian law.

In any case, the principles of necessity and proportionality that limit the use of force also apply to cyberspace; therefore, the attack and the response to it that only targeted "data" would have to be analyzed from that perspective (see Hongju Koh, Harold (2012), *International Law in Cyberspace*. In: Faculty Scholarship Series, Vol. 54, Yale Law School Faculty Scholarship).

Question 8. Is sovereignty a discrete rule of international law that prohibits States from engaging in specific cyber operations? If so, does that prohibition cover cyber operations that fall below the use of force threshold and which do not otherwise violate the duty of non-intervention?

Respect for the sovereignty of States is one of the fundamental pillars of international society. Indeed, United Nations General Assembly resolution 2625 (XXV), "Declaration on Principles of International Law concerning Friendly Relations and Co-operation

among States in accordance with the Charter of the United Nations,” states that sovereign equality includes the following elements:

- i. States are juridically equal;
- ii. Each State enjoys the rights inherent in full sovereignty;
- iii. Each State has the duty to respect the personality of other States;
- iv. The territorial integrity and political independence of the State are inviolable;
- v. Each State has the right freely to choose and develop its political, social, economic and cultural systems;
- vi. Each State has a duty to comply fully and in good faith with its international obligations and to live in peace with other States.

In relation to the threshold of use of force, for an action to be considered direct participation in hostilities, the resulting harm must reach a particular threshold. That threshold may be crossed when harm of a military nature is caused to a military objective, inflicting “death, injury, or destruction on persons or objects protected against direct attack” (see Melzer, Nils (2010), *Interpretive Guidance on the Notion of Direct Participation in Hostilities under International Humanitarian Law*, International Committee of the Red Cross (ICRC), Geneva, p. 47).

Question 9. Does due diligence qualify as a rule of international law that States must follow in exercising sovereignty over the information and communication technologies in their territory or under the control of their nationals?

In relation to the concept of due diligence, in its Advisory Opinion on the Legality of the Threat or Use of Nuclear Weapons (1996), the International Court of Justice said that it had become a part of generally accepted international law on the environment. In the Corfu Channel Case (1949), the same Court held that due diligence entailed the obligation that states not permit their respective territories to be used to prejudice the rights of other states by permitting activities that might produce such an outcome (see Brotons, Antonio Remiro (2010), *Derecho Internacional. Curso General*. Tirant Lo Blanch, Valencia, Spain, p. 434).

According to Annegret Bendiek, cybersecurity due diligence exists in cyberspace as an obligation for States in relation to the governance of processes and controls used to protect the information that flows through digital media. That obligation may also apply to other actors, such as international organizations and non-state actors (see Bendiek, Annegret (2016), *Due Diligence in Cyberspace. Guidelines for International and European Cyber Policy and Cybersecurity Policy*, Berlin, p.7)

Question 10. Are there other rules of international law that your government believes are important to highlight in assessing the regulation of cyber operations by States or actors for which a State is internationally responsible?

Although, at present, there is no international treaty in place on cyber warfare, the principles of customary international humanitarian law that are codified in the Geneva Conventions and their additional protocols—in other words, the principles of military necessity, humanity, proportionality, prohibition of superfluous injury and unnecessary suffering, and distinction—admit no exceptions in the case of armed conflict